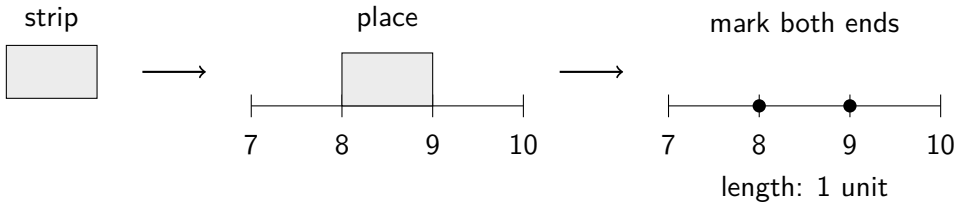
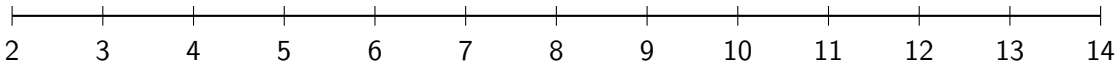
Use the four loose strips and the uncut row of squares. Each square is one unit. Lay strips along the ruler with both ends at printed marks.

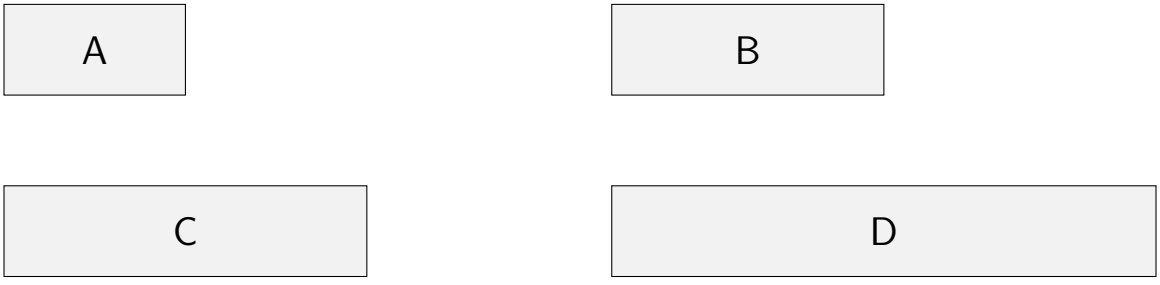


Problem 1: How many units long is each strip? Put each strip in different places on the ruler.



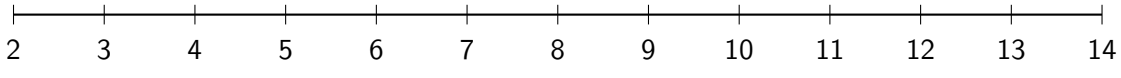
A: _____ B: _____ C: _____ D: _____



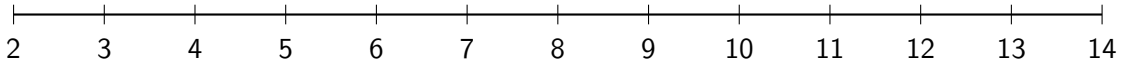


Problem 2: Find every place each strip can go with one end at 7. Draw its two ends on a ruler.

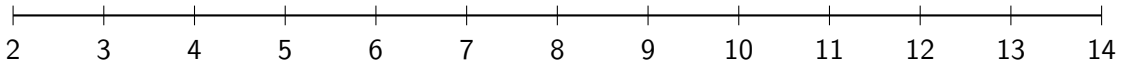
A



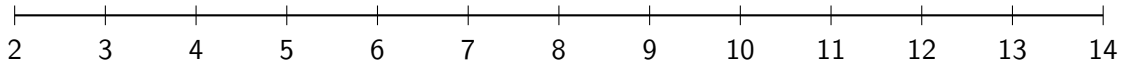
B



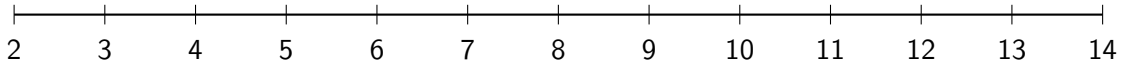
C



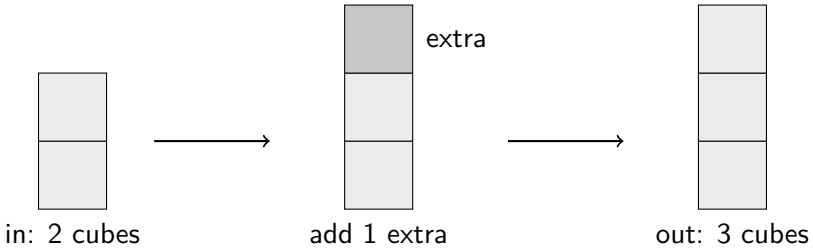
D



Problem 3: Which two strips can you put end to end and cover exactly with a third strip? Find every way.



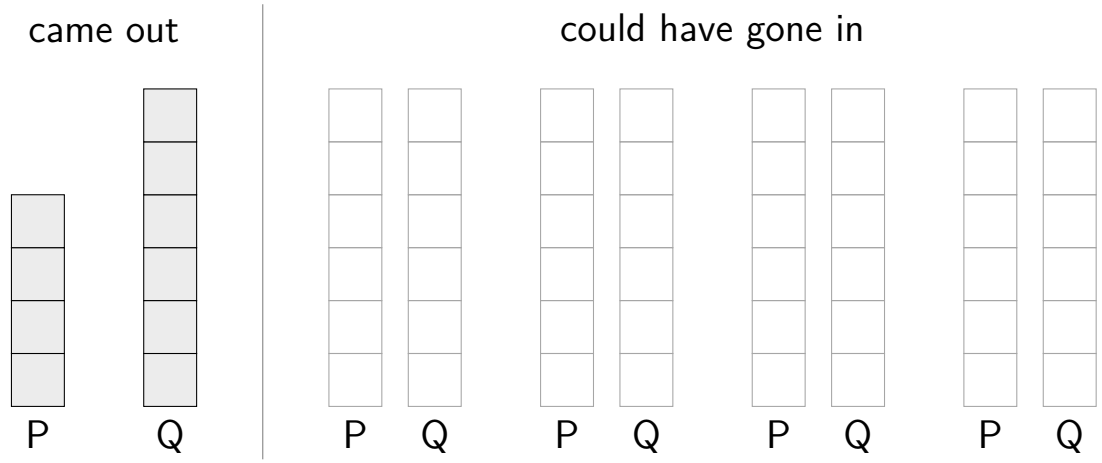
The machine takes a tower of cubes and gives back a tower with the same number of cubes plus a secret extra of 1, 2 or 3 cubes. The extra stays the same for the whole game.



Problem 4: Play the machine: choose a secret extra, and behind a cover build what comes out for each tower your partner gives you. Your partner finds your extra; then switch jobs.

Round	Towers in	Towers out	Extra
1			
2			
3			
4			

Problem 5: Build every pair of towers that could have gone in, and colour each pair you find. Could P be taller than Q?



Problem 6: P and Q still come out as 4 and 6 cubes, but each machine has its own extra. Build P and Q for each machine, and colour them.

